ResearchAgent



Research with a wallet.

An agent that answers from free sources first, then buys the one missing article from its writer, **over x402 on the XRP Ledger**, inside a budget you set. The writer is paid only by publishing the article's key on the ledger.

- XRPL Testnet · no real money
- Research writers are fictional · TEST-SGD stands in for RLUSD
- Runs on your machine · no hosted instance

— Progress since SingHacks

From a prototype a human approved, to one line that pays on Testnet

19–20 Sep
SingHacks
Prototype: a human approved every purchase.

10 Oct
AI Tinkerers
Live demo; first Testnet payments 7–9 Oct.

2–3 Nov
v0.1–v0.8
MCP server, Mac app, escrow, hire the agent, audit trail.

4 Nov
rc.1
Install without cloning.

5 Nov
rc.2
Real Testnet without cloning; key reveal by default.

7 Nov
main
`try`, TEST-SGD escrow, judge questions answered, Settings.

9–10 Nov
Lanes
`init` + publish for strangers; final acceptance.

12 Nov
`v1.0.0`
Release with the one-line judge path.

183
pull requests merged since SingHacks (21 Sep–7 Nov)

11
tagged releases, v0.1.0 → v1.0.0

1,293
TypeScript tests plus 258 Python tests on main, every hard gate included

224
shared receipt-conformance vectors; the TypeScript and Python verifiers agree on all

gh pr list --state merged · gh release list · PRs #381 (vectors), #387 (Python), #388 (TypeScript, check:fast 7 Nov) · docs/SFF-PUSH.md

One line, no clone: the whole judge path on real Testnet

```
$ npx -y …/releases/download/v1.0.0/researchagent-mcp-1.0.0.tgz try --testnet
```

```
Testnet · no real value · Fixture models and fictional writers
1. Create 9126E19D…B015B8CD
2. Finish · sha256(K) = Condition ✓ 805AA52F…1342EBB6
3. Refund · writer-signed BFAC5EF0…020585D1
4. Re-buy create 278087AB…0749871B
5. Re-buy finish ✓ 484A818A…AEB4C2EC8
6. Anchor 22081E19…A1E922F9
PASS testnet 12/12 (read-only, s.altnet)
RESULT PASS
Finished in 82.0 s · budget S$2.00 · spent S$0.70
```

Fresh npm cache, one Mac	rc.2	Now
try: install → six links → RESULT PASS	—	**84.2 s**
MCP package	~~21.4 MB~~	**3.4 MB**
MCP cold initialize	~~56.5 s~~	**6.3 s**
Local web app ready	~~134 s~~	**27.0 s**

> Measured 7 Nov with fresh faucet wallets. **Two of four cold attempts beat the 90 s deadline** (73.8 s, 84.2 s); two overran. If the faucet fails, it prints a labelled SIMULATED run and its reason.

PR #387 · docs/evidence/round4-j2/ (try-output.txt, try-timing.json, cold-timings.json, try-timeout*.json) · before = round-4 mock-judge observations on rc.2

— What we built

One question, one budget, and only code can spend it

1

Ask

A question and a budget, S$0 to S$5.

2

Confirm the plan

A 5-second sheet. It confirms the plan, not a purchase.

3

Search writers

Each writer returns abstracts and a signed fact menu, never premium text.

4

Decide

A decision model judges what is worth buying, weighted by trust.

5

Pay

Policy code locks payment in a key-reveal escrow on XRPL, within S$1 per source.

6

Verify, answer, anchor

Decrypt, check every promise, refund a broken one, cite, anchor the audit head.

Five hard gates, on every run

1. No premium bytes before a grant.
2. The budget is the only spending authority; only policy code can buy, never the LLM.
3. One charge per intent, across retries and restarts.
4. Every citation points at a delivered passage.
5. Anything simulated or substituted is labelled.

0 violations

Read-only gate audit: 10 runs, 568 observations, 48 fresh questions (3 Nov). Chaos: 200 runs with real SIGKILLs, 0 double charges (4 Nov). MCP spending caps default to zero; the calling model cannot raise them.

— Creativity · 20%

The writer is paid only by publishing the key: delivery and payment are one ledger event

Writer

Encrypt and promise

Fresh key K per buyer. Signs the quote: `sha256(K)`, ciphertext hash, passage manifest, price.

Buyer policy

EscrowCreate

Native XRPL Escrow. Condition = `sha256(K)`, with a CancelAfter deadline.

Writer

EscrowFinish

To be paid, the writer must reveal K as the fulfilment. K is now public on the ledger.

Buyer policy

Decrypt and check

AES-256-GCM decrypt, check every promised passage, then grant and cite.

Promise kept	Create	→ Finish, K revealed	→ decrypt ✓ passages ✓	→ cited	`278087AB…` →	`484A818A…`
Promise broken	Create	→ Finish, K revealed	→ claim missing	→ writer-signed refund + fraud proof	`805AA52F…` →	`BFAC5EF0…`
Writer no-show	Create	→ no key by CancelAfter	→ EscrowCancel, principal back, nothing granted		`8DA4ECE7…` →	`F89DD9B1…`

> **What the ledger does and doesn't enforce.** Key reveal makes **delivery** atomic: no key, no payment. **Quality** rests on the signed promises, a fraud proof anyone can check, and reputation. After a finish, the refund is writer-signed: our fixture writer cooperates, and a real writer could refuse. No smart contract: Escrow with a PREIMAGE-SHA-256 condition is native XRPL.

The same key-reveal escrow, settled in a stablecoin-style token

TEST-SGD through XLS-85 token escrow

- **Opt in:** add `--iou` to the zero-clone MCP or web app. S$1 maps to 1 TEST-SGD.
- **Preflight refuses** a missing issuer locking permission, frozen or unauthorized trust lines, short balances and transfer fees.
- **Both verifiers** check currency, issuer and the writer's trust-line balance change, since EscrowFinish has no delivered_amount: TypeScript and Python PASS 6/6.
- **Testnet runs:** three purchases in 15.0–18.4 s; a no-show cancel returned the principal in 132.3 s.

`create E36AF9F1…18878C5C · finish 384CA34F…0181C02F5 · anchor 8174E3BE…ADC35A3`

Official RLUSD: payments yes, escrow one flag away

- **Payments work.** On 3 Nov, official Testnet RLUSD paid a writer 0.90 (`1F2449A4…`) and a broken promise was refunded 0.30.
- **Escrow is blocked by the issuer, not the network.** TokenEscrow is enabled on Testnet, but the RLUSD issuer's flags (2174353408) lack AllowTrustLineLocking. One AccountSet by the issuer would allow it.
- **So we escrow our own token** and say so on every amount, receipt and ledger row. It is our top builder-feedback item.

TEST-SGD · IOU issued by our Testnet issuer · no real value · stand-in for RLUSD

No peg, backing or mainnet claim

PRs #381, #384 · docs/evidence/m1-iou/, round3-e/runs.json, rlusd.md · BUILDER-FEEDBACK §1 (issuer flags at ledger 22058675; refusal predicted, never submitted)

— Technical depth · 20%

Don’t trust our agent. Verify it, against the ledger.

```
$ pip install …/researchagent-1.0.0…whl
$ researchagent verify UC3.receipts.json --ledger
PASS checks: 57 PASS, 0 FAIL, 2 SKIP
PASS audit anchor: Validated XRPL Testnet ledger 22057921
PASS testnet 12/12 (read-only, s.altnet)
RESULT PASS
```

Python, standard library only, no Node. Any receipt the app exports (Export ⋯ → receipts.json) can be checked the same way.

- **Two independent verifiers**, TypeScript and Python, agree on all 224 shared vectors, including tampered amounts, issuers, refunds and a missing refund.
- **45/45 Testnet transactions re-validated live** on our public ledger page in 1.97 s, read-only.
- **Every paid Testnet answer is anchored:** an AccountSet whose memo is the audit-chain head, 12 drops. No purchase, no anchor.
- **Persist before submit.** A restart reconciles the same signed hash and never signs twice; a missing transaction is *pending*, not failed.
- **Exact delivery.** Partial payments rejected before submit; delivered amount, currency and issuer checked after.
- **Prompt injection can’t spend:** model inputs are capped and untrusted, and purchases come only from policy code.

PRs #377, #381, #387, #388 · docs/evidence/round4-j2/python-verifier.txt (README UC3 evidence) · round4-l4/live-ledger.json · BUILDER-FEEDBACK top five

— Reachability · 25%

Every way in runs from one package, for people and for other agents

One-line judge path

`try --testnet` buys, refunds, re-buys, anchors and verifies, then exits.

84.2 s from an empty cache · six links · RESULT PASS

Claude Code, any MCP client

`claude mcp add … --testnet`. Seven tools; operator caps default to zero.

Cold initialize 6.3 s · `--iou` for TEST-SGD

Local web app, no clone

`app --testnet` serves the real UI on 127.0.0.1 with faucet wallets. Ask your own question.

Ready in 27.0 s from an empty cache

SDKs: TypeScript and Python

Ask, stream, verify. No purchase method: only backend policy can buy.

Python verifier needs no Node

Writers in one command

`publish ./notes --testnet`: faucet wallet, XRPL Credential, key-reveal escrow.

Credential in 43.8 s · first try from `init`: `33.9 s`

Hire the agent over x402

Another agent pays ResearchAgent; it buys, pays a royalty to every cited writer and returns the change.

Five-hop chain on Testnet, 3 Nov

> **XRPL questions · fixture models · local candidate:** team-authored real build notes, not independent. **Held-out 8/20 · dev 16/20** bought and cited; 0 blanks / gate violations each. Online receipts 8/20 held-out, 16/20 dev: abstentions lack anchors. **Local only:** no hosted instance (`{{REGISTRY_STATUS}}`).

PRs #376, #378, #380, #384, #387, #390 · docs/evidence/round4-j2/, j5-coverage/, round3-c/, hire-274.md · J3 lane pending merge

— Feasibility · 20%

Fractions of a cent in fees, and it recovers from failure

435 drops

network fees per key-reveal purchase

12 create + 423 conditional finish, measured on Testnet

S$0.00087

fees per purchase, projected

At an assumed S$2.00 per XRP; a projection, not a quote

US$0.0026

model cost per question

Live decision and coverage models, 48 fresh questions, 3 Nov

894 drops

all six UC3 transactions

Two buys, a refund and an anchor; anchor alone 12 drops

43.9 s

nightly Testnet purchase in CI

Devcontainer boots and buys on real Testnet; green 6 and 7 Nov

0

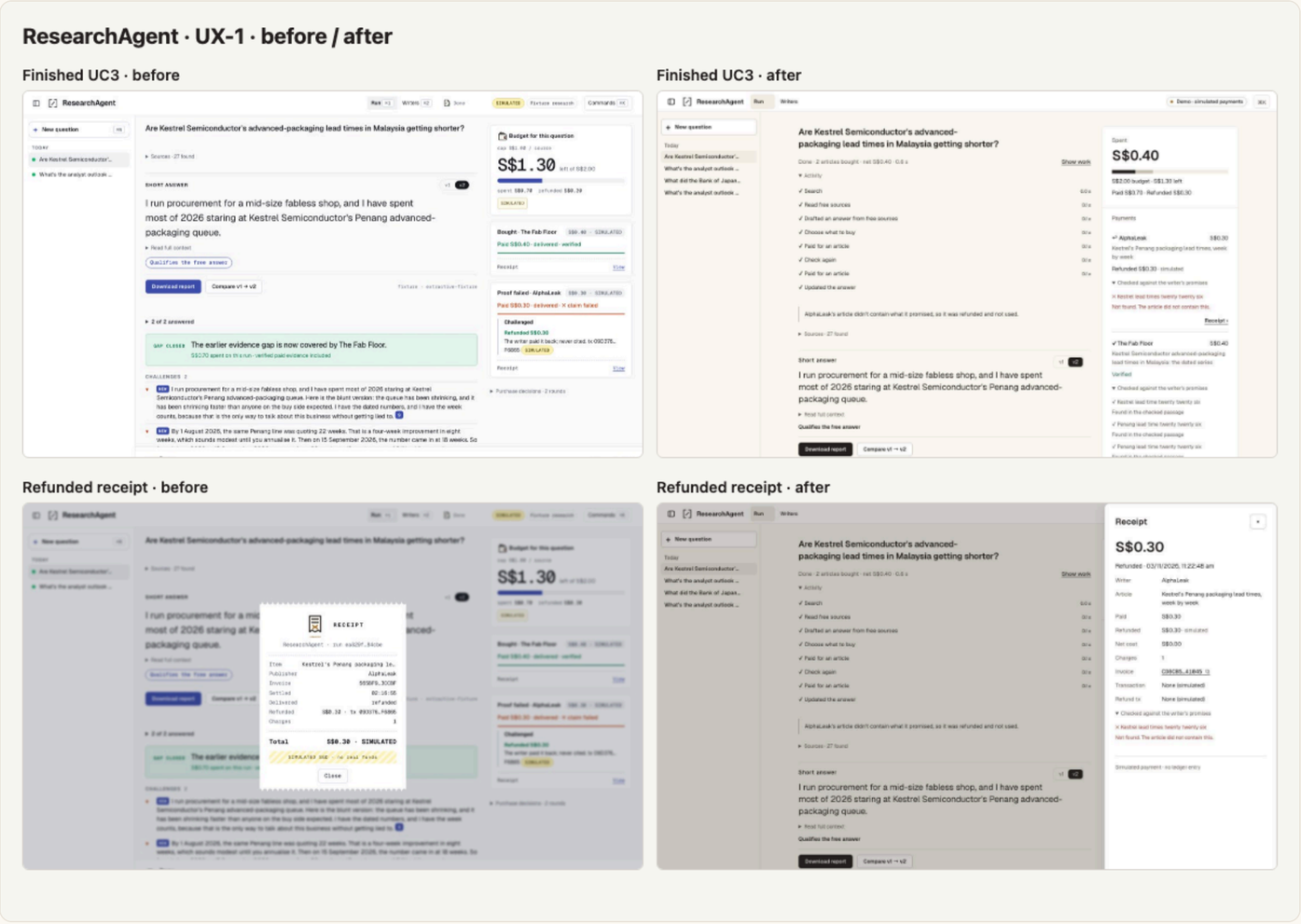
double charges under chaos

200 runs with real SIGKILLs; reconciler recovers in-flight payments

> **Does it buy the right things?** On 48 fresh synthetic questions, **93.15%** of gross article spend bought a requested fact retained in the final cited answer (95% CI 81.25–100.00%); wasted spend 6.85% (0.00–18.75%). Live decision models, fixture research, simulated payments, 3 Nov. The gain over the previous policy (+3.08 pts, 0.00–9.43) is not significant; the labels are same-vendor and await human review.

README cost card (round3-a) · round4-l4/judge-path.md · eval/scorecard/fresh-results.json · FRESH-AUDIT.md · codespace-testnet.yml · PRs #376, #381

The money is the hero, and every label tells the truth



- **One environment indicator** in the header: “Testnet · no real value” or “Simulated”.
- **Payment rows change state in place:** escrow created, key published, sha256(K) = Condition ✓, refunded. Each opens its receipt and hash.
- **A refund says why and who signed:** “Refund · writer-signed · fixture writer cooperates (a real writer could refuse).”
- **Proof stays out of the way:** “6/6 Testnet hashes verified live ✓ · show” expands on demand.
- **Motion only for money**; the pixel agent marks loading states only and never stands in for a receipt.
- **One Settings page** for the web app and the Mac app: nine sections; changes to caps and keys are guarded and audited.

Image: release v0.5.0 before-after.png (offline demo, simulated payments) · PRs #379, #385, #386, #388 · docs/design/UX-DIRECTION.md

Ranked builder feedback for Ripple, with runnable repros

1 **Issuer permission blocks RLUSD escrow**

TokenEscrow is enabled, but the issuer lacks AllowTrustLineLocking. Fix: a network and issuer preflight, with dated flags.

2 **An issued amount must mean the delivered amount**

Amount alone can’t prove exact delivery, and EscrowFinish has no delivered_amount. Fix: negative issued-asset vectors.

3 **Public endpoints disagree on retained history**

Unavailable history must not read as failed settlement; HTTP lacks browser CORS. Fix: document retention and archival lookups.

4 **Escrow expiry doesn’t return funds by itself**

A failed-proof escrow stayed locked past its deadline until an explicit EscrowCancel. Fix: document expiry vs cancellation.

5 **New: a signed IOU quote can overflow the 1 KiB Memo**

Observed 7 Nov on the TEST-SGD path. Fix: check serialized Memo size before signing and return the reason.

\+ **Mainnet readiness: a gap checklist, not a claim**

Key custody (KMS or external signer), FX quotes, facilitator SLA, KYB and sanctions screening.

> **7** ranked findings with repro scripts · **29** dated implementation-log entries (B01–B29) · **15** September submissions · **7** upstream issue drafts with a filing kit. Filed upstream: `{{UPSTREAM_FILED_URLS}}`.

— Check it yourself

Every claim has a Testnet hash or a command

What	Where
Run the judge path	`npx -y … try --testnet`
Release, every artifact	`v1.0.0` on GitHub Releases
Replay of a Testnet run	…/theFastandtheFungible/demo/
45 hashes, checked live	…/theFastandtheFungible/ledger/
Demo video	`{{VIDEO_URL}}`
Code and evidence	github.com/Collaboration95/…

Before mainnet

- Payer keys in a KMS or external signer
- RLUSD escrow once the issuer allows it; live FX
- KYB and a real sanctions-screening provider
- Independent writers with consent (none yet)
- Signed, notarised Mac app

